Lecture 12: Interpret the Dual Problems

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Announcement

- Homework 3 due next today
- Homework 4 posted, due next Wednesday (3/15)

Recap: Duality

What have we learned so far about LP duality?

- Given any linear program, we can construct its dual
- Duality theorems (suppose the primal problem is a minimization problem)

Theorem (Weak Duality Theorem)

If $\mathbf{x}$ *is a feasible solution to the primal problem and* $\mathbf{y}$ *is a feasible solution to the dual problem, then*

$$\mathbf{b}^T\mathbf{y} \leq \mathbf{c}^T\mathbf{x}$$

- Primal (dual) feasible solution can be used to construct bounds for the objective value of the dual (primal) problem
- If one problem is unbounded, the other one must be infeasible
- If $\mathbf{x}$ is primal feasible, $\mathbf{y}$ is dual feasible, and $\mathbf{c}^T\mathbf{x} = \mathbf{b}^T\mathbf{y}$, then they are both optimal — Optimality Conditions

Strong Duality Theorem

Theorem (Strong Duality Theorem)

If a linear program has an optimal solution, so does its dual, and the optimal value of the primal and dual problems are equal

- In the proof of the strong duality theorem, we showed that the simplex method can actually find the dual optimal solution when it finishes
- We can also find the dual optimal solution in the final simplex tableau in certain circumstances

All possible states of a pair of primal/dual linear programs

	Finite Optimum	Unbounded	Infeasible
Finite Optimum	$\surd$		
Unbounded			$\surd$
Infeasible		$\surd$	$\surd$

Interpret the Dual Problem

In the following, we will look at some examples and provide interpretations to the dual problems, and see how the duality theory can help us solve problems

- The production planning problem
- The multi-firm alliance problem
- The alternative systems problem
- The maximum flow problem

Production Planning Problem

Recall the production planning problem:

$$\begin{array}{lrrl} \text{maximize} & x_1 & +2x_2 & \\ \text{subject to} & x_1 & & \leq 100 \\ & & 2x_2 & \leq 200 \\ & x_1 & +x_2 & \leq 150 \\ & x_1, & x_2 & \geq 0 \end{array}$$

The objective corresponds to profits; the constraints are resource constraints.

Dual of the Production Planning Problem

We write down its dual and associate the three constraints with dual variables p_1, p_2 and p_3.

$$\begin{array}{rrrrrl} \text{minimize} & 100p_1 & +200p_2 & +150p_3 & & \\ \text{subject to} & p_1 & & +p_3 & \geq & 1 \\ & & 2p_2 & +p_3 & \geq & 2 \\ & p_1, & p_2, & p_3 & \geq & 0 \end{array}$$

What is the meaning of this optimization problem?

- Suppose someone wants to buy all resources of the firm at unit prices p_1, p_2 and p_3. What prices he should offer?
- Since one unit of resource 1 and resource 3 can produce one unit of product 1 which worths \$1. Therefore, if $p_1 + p_3 < 1$, the firm would rather produce by itself (than agree to sell).
- Similarly, it wouldn't sell if $2p_2 + p_3 < 2$.
- Other than that, the buyer wants to buy the resources for the lowest price, and the dual problem finds that price.

Discussions

In this example, we interpret the dual variables to be the fair price of each resource

- This is a very common type of interpretation for dual variables
- In the next lecture we will see why this makes sense
- Below we look at an extension of the production planning problem and see how the dual problem can provide help for solving an important question

Multi-Firm Alliance Problem

Suppose there is a collection of firms $1, 2, ..., m$, each making the same set of products.

- Firm i has resource $\mathbf{b}_i$ and has consumption matrix A and profit $\mathbf{c}$ for each product (A and $\mathbf{c}$ are the same across all firms)

Therefore, each firm is trying to solve a production problem:

$$\begin{aligned} \text{maximize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b}_i \\ & \mathbf{x} \geq 0 \end{aligned}$$

Denote the optimal profit for firm i by V^i

Multi-Firm Alliance Problem

Now suppose the firms can form an *alliance* so that they can *pool* their resources and produce together.

Assume a subset of firms S form an alliance, then to maximize the profit of the alliance S, we solve the following LP:

$$\begin{aligned} \text{maximize}_{\mathbf{x}} \quad & \mathbf{c}^T\mathbf{x} \\ \text{s.t.} \quad & A\mathbf{x} \leq \sum_{i\in S} \mathbf{b}_i \\ & \mathbf{x} \geq 0 \end{aligned}$$

The optimal profit for the alliance is denoted by V^S.

We call the alliance involving all firms the *grand alliance*. In a grand alliance, all resources are pooled together. We denote the optimal profit for the grand alliance by V^*.

Multi-Firm Alliance Problem

One important question before forming an alliance is:

- How to allocate the profit of the alliance to each of its members so that each member/suballiance has the incentive to stay in that alliance?

Now let's consider the grand alliance. Our question is whether the firms have incentive to form a grand alliance and what allocation rule we should use?

What constraints do we have?

1. The total allocation equals the total profit
2. The allocated profits to any suballiance should be greater than what the suballiance could get if they separate from the grand alliance (no subset of players would want to leave the alliance)

Multi-Firm Alliance Problem

In order to make the grand alliance stable, an allocation $(z_1, ..., z_m)$ must satisfy

$$\sum_{i=1}^{m} z_i = V^*$$
$$\sum_{i \in S} z_i \geq V^S, \quad \forall S \subseteq \{1, ..., m\}$$

We call such z_i's the *core* of the grand alliance.

- It is not trivial to find such z_i's or even prove its existence
- It involves checking 2^m inequalities
- However, we can use duality to find such a solution easily

Multi-Firm Alliance Problem

The solution: consider the dual of the grand alliance production problem:

$$\begin{aligned} \text{minimize}_{\mathbf{y}} \quad & \left(\sum_{i=1}^{m} \mathbf{b}_i\right)^T \mathbf{y} \\ \text{s.t.} \quad & A^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq 0 \end{aligned}$$

Denote the optimal solution by $\mathbf{y}^*$

Theorem

Define $z_i = \mathbf{b}_i^T \mathbf{y}^*$. Then $(z_1, ..., z_m)$ is in the core of the grand alliance.

- Basically, $\mathbf{y}^*$ can be viewed as the prices for the resource and we just allocate according to the value of resource of each firm

Proof

First, by the strong duality theorem, we know that $\sum_{i=1}^{m} z_i = V^*$. Now we only need to show that for any subset S,

$$\sum_{i \in S} z_i \geq V^S.$$

Consider the production problem for alliance S and its dual:

Primal	$\max_{\mathbf{x}}$	$\mathbf{c}^T\mathbf{x}$	$\Big\|$	**Dual**	$\min_{\mathbf{y}}$	$\left(\sum_{i \in S} \mathbf{b}_i\right)^T \mathbf{y}$
	s.t.	$A\mathbf{x} \leq \sum_{i \in S} \mathbf{b}_i$			s.t.	$A^T\mathbf{y} \geq \mathbf{c}$
		$\mathbf{x} \geq 0$				$\mathbf{y} \geq 0$

Note that $\mathbf{y}^*$ is still a feasible solution to the dual. Therefore, by the weak duality theorem,

$$\sum_{i \in S} z_i = (\sum_{i \in S} \mathbf{b}_i)^T \mathbf{y}^* \geq V^S$$

Alternative Systems

Given a set of linear inequalities:

$$A^T \mathbf{y} \leq \mathbf{c}$$

An important question is: whether the system has a solution?

- It is easy to verify that it has a solution, one only needs to find a solution (we call it a *certificate*)
- To disprove the existence, can we also have such a certificate?

The answer: Yes.

- If we can find a vector $\mathbf{x}$ satisfying

$$A\mathbf{x} = 0, \quad \mathbf{x} \geq 0, \quad \mathbf{c}^T \mathbf{x} < 0$$

Then there must be no solution to the system $A^T \mathbf{y} \leq \mathbf{c}$

Proof. Consider the pair of primal-dual linear program:

Primal		**Dual**	
min	$\mathbf{c}^T\mathbf{x}$	max	0
s.t.	$A\mathbf{x} = 0, \mathbf{x} \geq 0$	s.t.	$A^T\mathbf{y} \leq \mathbf{c}$

If there exists $\mathbf{x}$ such that

$$A\mathbf{x} = 0, \quad \mathbf{x} \geq 0, \quad \mathbf{c}^T\mathbf{x} < 0$$

Then by scaling this $\mathbf{x}$, we can make $\mathbf{c}^T\mathbf{x}$ arbitrarily negative, therefore, the primal problem must be unbounded.

However, for an LP, if the primal problem is unbounded, then the dual problem must be infeasible. Thus such an $\mathbf{x}$ is a *certificate* that the linear system $A^T\mathbf{y} \leq \mathbf{c}$ is infeasible. □

Alternative Systems

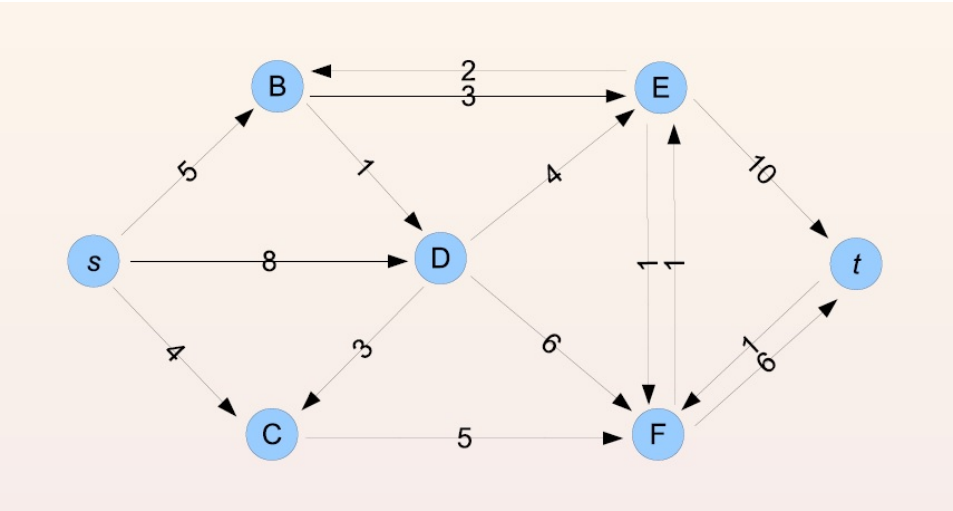
One can construct many more pairs of such alternative systems.

- It is hard to directly prove something is not feasible
- LP duality provides an alternative approach, transforming the problem to proving something is feasible

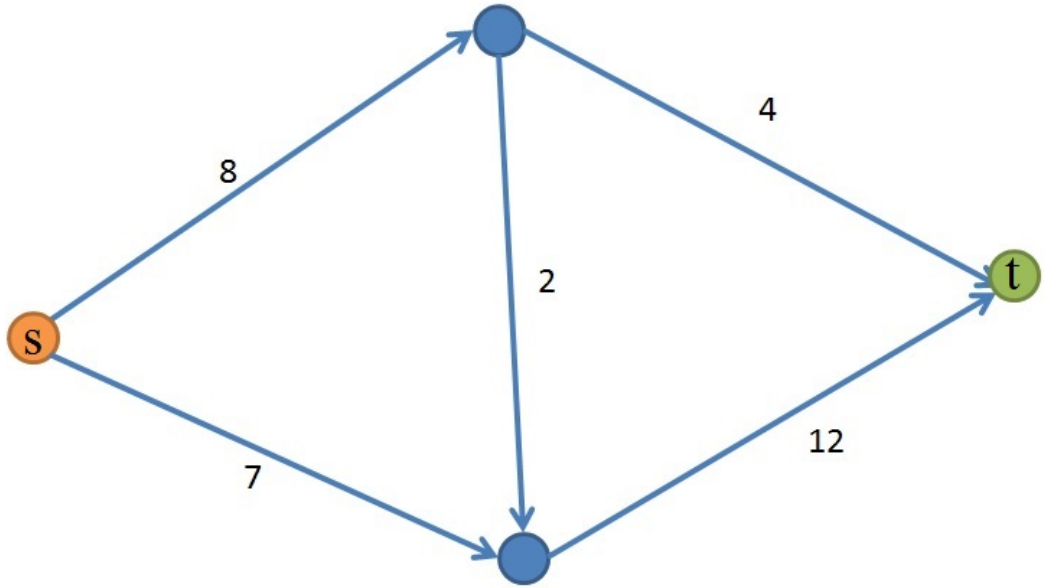
Maximum Flow Problem

The maximum flow problem can be described as follows:

- Given a directed, weighted graph $G = (V, E)$ and a pair of nodes s and t (V is the set of nodes, E is the set of edges)
- One can think this as a traffic network
- There is an edge capacity w_{ij} on each edge
- Question: What is the largest amount of flow one can send from s to t, subject to the capacity constraints?



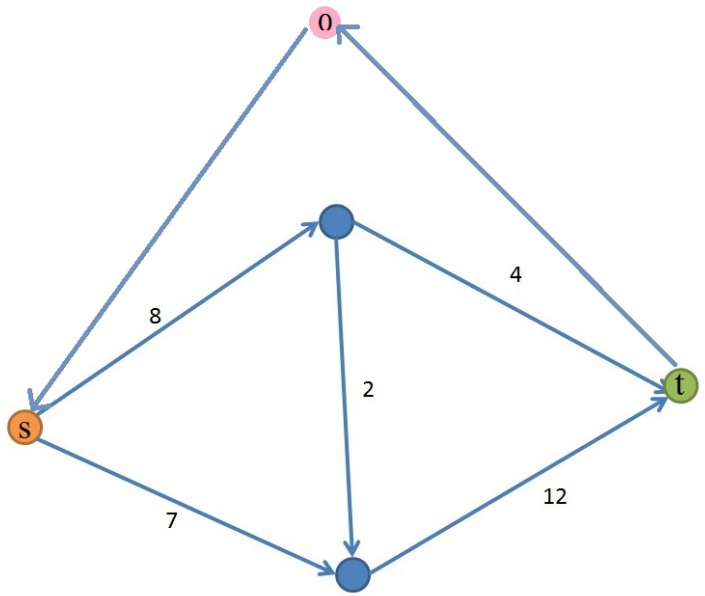
A Concrete Example



One Transformation

Assume there is an imaginary node o, with edges (o, s) and (t, o). There is no capacity constraint on those two edges

- The problem becomes a closed system. One wants to maximize the flow from o to s, which we denote by Δ.



LP Formulation

Using this transformation, we can write down the LP formulation. Let x_{ij} denote the amount of flow across edge (i, j).

$$
\begin{aligned}
\text{maximize}_{\mathbf{x},\Delta} \quad & \Delta \\
\text{subject to} \quad & \textstyle\sum_{j:(j,i)\in E} x_{ji} - \sum_{j:(i,j)\in E} x_{ij} = 0, && \forall i \neq s, t \\
& \textstyle\sum_{j:(j,s)\in E} x_{js} - \sum_{j:(s,j)\in E} x_{sj} + \Delta = 0 \\
& \textstyle\sum_{j:(j,t)\in E} x_{jt} - \sum_{j:(t,j)\in E} x_{tj} - \Delta = 0 \\
& x_{ij} \leq w_{ij}, && \forall (i,j) \in E \\
& x_{ij} \geq 0, && \forall (i,j) \in E
\end{aligned}
$$

- The first constraint is the flow balancing constraints for all nodes other than s and t
- The second (third, resp.) constraint is the flow balancing constraints for node s (t, resp.)

Dual of the Maximum Flow Problem

We construct the dual problem:

$$\begin{aligned} \text{minimize} \quad & \sum_{(i,j)\in E} w_{ij} z_{ij} \\ \text{subject to} \quad & z_{ij} \geq y_i - y_j, \qquad \forall (i,j) \in E \\ & y_s - y_t = 1 \\ & z_{ij} \geq 0 \end{aligned}$$

What does the dual problem mean?

First assume all y's are 0 or 1. Then

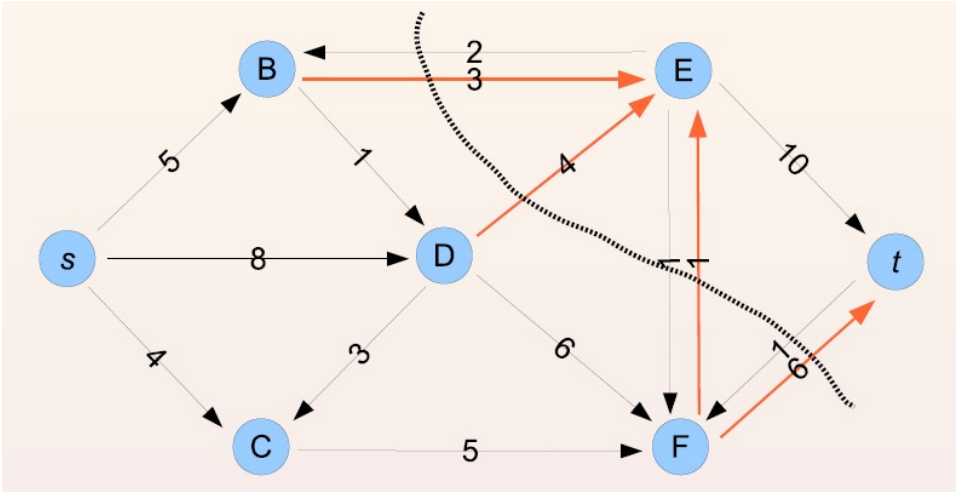
- We assign a label (0 or 1) to each node, 1 to s and 0 to t.
- If i has a larger label than j for $(i,j) \in E$, there is a cost w_{ij}.

Interpretation of the Dual

The dual problem is equivalent to finding a subset S of vertices containing s but not t, that minimizes the weight of the cut, i.e.

$$\sum_{i \in S, j \notin S} w_{ij}$$

- This is called the min-cut problem



Strong Duality Says..

Theorem

The maximum flow of the network is equal to the smallest cut size of any subset S of vertices.

Corollary

If the value of a flow is equal to the value of some cut, then both are optimal.

- One can view the min-cut as the bottleneck of the network.
- The maximum flow that can be sent through this network is equal to the tightest bottleneck of this network.

This is one classical example of dual problems: maximum flow versus minimum cut.